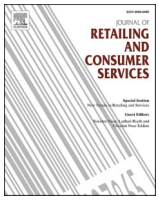
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E-commerce enterprise flexibility leading to better customer perception

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ABSTRACT

This study aims to assess the level of organizational flexibility in e-commerce enterprises and to develop a structural model that identifies the key factors contributing to flexibility and their impact on customer service perception. The importance of customer perception in e-commerce is undeniable; however, the intra-organizational factors that contribute to it remain unclear. One such factor is flexibility, which has not yet been thoroughly researched. The survey was conducted on a random sample of 213 e-commerce companies of different sizes. The structural equation modelling (SEM) revealed that two factors significantly influence customer perception: assortment flexibility along with price and margin flexibility. Additionally, it was found that management's openness to change and the type of software utilized affect the degree of these flexibilities. This indicates that e-commerce companies should implement strategies to enhance flexibility in shaping their offerings to improve customers' perception of the services provided. It is recommended to foster management's openness to change in order to respond swiftly to evolving customer expectations. Furthermore, software functionalities that facilitate greater organizational flexibility are highly valued.

1. Introduction

Dynamic changes in the market significantly impact the commercial activities of businesses, including e-commerce companies, whose market share continues to grow (Nayyar et al., 2023). Recently, the surge in interest in online shopping has been strongly influenced by the Covid-19 pandemic. Many consumers perceive online shopping as a safer alternative, providing individuals in quarantine or isolation the opportunity to make purchases without leaving their homes (Podorova-Anikina et al., 2022; Zhang et al., 2023). Simultaneously, a shift in consumers' needs and expectations regarding shopping and related services is perceptible. Interest in purchasing certain products is increasing, e.g., food, medicine, and dietary supplements, while interest in others is declining, e.g., furniture, home accessories (Lone and Weltevreden, 2022; Lone and Weltevreden, 2023; Weltevreden, 2024). Purchasing decisions are influenced by various factors; however, it is clear that customers are increasingly considering the quality, eco-friendliness, and origin of products when adding them to their shopping carts (Buraczyńska et al., 2022; Štofejová et al., 2023). Nevertheless, consumers' purchasing needs are not static, and trends in these changes can vary. Additionally, each customer may have unique expectations regarding the terms of purchase, which are also influenced by their perception of service quality. Consequently, e-commerce companies must continuously identify and adapt to their customers' evolving needs to remain competitive.

In a dynamic environment, the advantages of organizational flexibility in exploring new opportunities while leveraging existing products and services are clear (De la Gala-Velásquez et al., 2023; Pryor et al., 2022; Saeed et al., 2021). Organizational flexibility is a complementary capability that enhances the effectiveness and benefits of other organizational competencies (Joseph Jerome et al., 2024; Srinivasan and Swink, 2018). It is regarded as a crucial organizational capability that amplifies the impact and advantages of other capabilities (Jain et al., 2020; Joseph Jerome et al., 2024). Furthermore, organizational flexibility plays a vital role in management decision-making concerning enterprise operations (Joseph Jerome et al., 2024; Shukla et al., 2019).

Research on organizational flexibility has demonstrated that companies possessing this characteristic are better equipped to maintain competitiveness and enhance profitability in various market environments (Côrte-Real et al., 2020; Joseph Jerome et al., 2024). A study conducted within the fashion industry revealed a significant positive impact of organizational flexibility, indicating a direct influence on supply chain agility and an indirect effect on overall company performance (Chan et al., 2017; Joseph Jerome et al., 2024). However, few

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studies have explored the relationship between the flexibility of e-commerce companies and customer perception of service quality and satisfaction (Urban and Buraczyńska, 2017). These studies generally suggest that the integration of devices and systems, along with their flexibility, enhances customer satisfaction. However, they also indicate that devices and systems, along with their flexibility, enhance customer satisfaction. Nevertheless, infrastructure alone cannot directly or independently improve customer satisfaction (Yu et al., 2015).

Hence, this study aims to assess the level of organizational flexibility in e-commerce enterprises and to develop a structural model that illustrates the key factors essential for achieving flexibility. Additionally, it will explore how flexibility influences customer perception of service quality. The study seeks to identify critical factors that determine organizational flexibility and propose strategies for enhancing the enterprise's flexibility.

Despite the increasing significance of organizational flexibility in business, particularly in the digital economy, there remains a limited understanding of how the flexibility of digital service organizations influences customers' perception of the services they receive. Several issues remain unclear, including the components of e-commerce flexibility, the factors that contribute to greater flexibility, and the interdependence of internal and external factors. This study employs a quantitative research strategy, which allows for the investigation of measurable issues and the development of a model regarding flexibility in e-commerce organizations. The novelty of this study lies in its combined focus on internal organizational factors and external customer perception, as well as its integration with external customer quality assessments. Together, these elements provide new insights into the relationship between organizational flexibility and the quality of e-commerce services.

2. Theoretical framework

2.1. Flexibility in commerce and e-commerce

Flexibility in commerce is a multidimensional concept that is approached from various perspectives, such as strategic flexibility, operational flexibility, organizational flexibility or innovation flexibility (Brozović et al., 2023; Sushil, 2001). Researchers offer a range of definitions for organizational flexibility, highlighting the significance of its diverse dimensions. Organizational flexibility is defined as the capacity of an organization to modify its characteristics by effectively utilizing and reallocating both tangible and intangible resources (De la Gala-Velásquez et al., 2023; Yousaf and Majid, 2018). The flexibility serves as a response to a changing environment and is evident in the ability to reallocate and reconfigure the enterprise's resources, as well as its strategic and operational processes (De la Gala-Velásquez et al., 2023; Singh et al., 2021). Agility encompasses the ability to detect temporary, short-term changes and respond swiftly, as well as the capacity to recognize fundamental, long-term changes and adapt by adjusting the supply chain configuration or engaging in innovation. It has been measured using indicators such as customer delivery times and replacement times for purchases (Eckstein et al., 2015; Swafford et al., 2008; Swafford et al., 2006). In other studies, the aspect of organizational flexibility has been assessed primarily through changes in organizational structure (De la Gala-Velásquez et al., 2023; Dubey et al., 2019; Srinivasan and Swink, 2018).

Researchers studying organizational flexibility emphasize its various forms. One context involves creating an offer, where flexibility is defined as the company's ability to adjust its product assortment to maximize profit (Okręglicka et al., 2021). An important factor to consider is the range of products offered (He et al., 2019), as well as the pricing and margin levels (Yu, 2015). The research also modelled the flexibility of supplier selection (Ware et al., 2014). Supplier flexibility is crucial for customers, as it encompasses the ability to adapt to changing buyer needs regarding delivery schedules, order quantities, and product modifications (Chirra and Kumar, 2018; Üstündağ and Ungan, 2020). Additionally, it includes production flexibility, which considers suppliers' experience; demand flexibility, which pertains to order processing; and a variety of flexibilities relating to packaging (Van der Rhee et al., 2009).

Human resources is another area in which organizational flexibility is examined. This concept encompasses the adaptability of human resources, defined broadly as the extent to which these resources enable the organization to respond to changing circumstances through the number of employees and their ability to perform diverse functions (Mgbemena-Nsofor et al., 2024). Consequently, the company offers employees a flexibly agreed scope of work and they benefit from flexible working conditions (e.g., shortened working week, flexible work schedules and working hours, and hybrid work) (Coenen and Kok, 2014; Giannikis and Mihail, 2011), which is reflected in their productivity and the company's performance (Okręglicka et al., 2021; Rastogi et al., 2016). Additional practices may include recruiting employees on either a permanent or temporary basis for specific tasks, as well as outsourcing and engaging individuals who can manage various tasks (Ozer, 2002).

Another field of organizational flexibility is technology, with a particular emphasis on information technology within the e-commerce sector. Information technology flexibility is a broad concept in the enterprise. It encompasses the organization's IT hardware, software, and network systems, focusing on their scalability and adaptability (Sukur, 2020; Tallon and Pinsonneault, 2011). Researchers indicate that technological flexibility involves software compatibility and the ability to adapt to the diverse technologies utilized by buyers and sellers. The use of flexible technologies is intended to meet the growing and changing requirements of its customers. The way e-commerce companies acquire and use technology can also be flexible (Ozer, 2002).

In the context of services, the authors distinguish delivery flexibility (the ability to alter the order details or its date), volume flexibility (the ability of the company to adjust the total production volume), and mixed flexibility (the ability to modify various products within a specified timeframe while incurring minimal additional costs) (Okręglicka et al., 2021). For enterprises providing electronic services, the flexibility of e-services is understood to manifest across all phases of the customer activity cycle. In each phase, a distinct type of flexibility is deemed dominant, i.e. the most important for customers. Specifically, information flexibility is paramount during the pre-purchase phase, customization flexibility during the purchase phase, and customer support flexibility during the post-purchase phase. Research has shown that companies offering highly flexible e-service provide customers with up-to-date product and service knowledge, offer more customized products or services, and provide more communication options to solve post-purchase problems. Consequently, customer readiness is considered a moderator of the relationship between internal flexibility and firm performance (Jin and Oriaku, 2013). Therefore, it can be inferred that flexibility determines customer readiness and impacts organizational performance. Additionally, other studies indicate that flexibility exerts a significant and positive effect on customers' perceived value, which, in turn, significantly and positively affects their levels of satisfaction (Nursyirwan et al., 2020).

Based on the literature, it can be inferred that:

Supposition 1: The flexibility of an e-commerce company influences customers' perception of its services.

2.2. Determinants of organisational flexibility

Previous research has consistently addressed the issue of flexibility, its determinants, and its impact on a company's performance and competitiveness (e.g., De la Gala-Velásquez et al., 2023; Rubio-Andrés and Gutiérrez-Broncano, 2014). Studies have explored the relationship between enterprise information systems and their strategic flexibility. The literature suggests that enterprise information systems within organizations influence information flexibility. Specifically, there is a

positive impact on analytical flexibility and a negative impact on reporting flexibility (MacKinnon et al., 2008). In line with contingency theory, research has examined the interplay between organizational flexibility and management support, which has enabled companies to develop innovative solutions in response to market changes (De la Gala-Velásquez et al., 2023; Hambrick and Mason, 1984). Contingency theory posits that the most effective management approach is contingent upon various internal and external factors, including the organizational environment and structure. In this context, it underscores the necessity for organizational flexibility and adaptive management strategies to address the specific challenges that a company encounters (Fiedler, 1964).

The literature also discusses flexible internal competencies, including logistics flexibility and IT application flexibility. It has been suggested that companies with high levels of internal flexibility, particularly in logistics and information technology, are more likely to offer a higher level of flexible e-services (Jin and Oriaku, 2013). Organizations with a flexible IT infrastructure can gain a competitive advantage, as this flexibility supports e-business applications in the design, development, and implementation of diverse solutions. It enables managers to effectively control the company's external environment and respond swiftly to competitors' strategic moves (Byrd and Turner, 2001). Furthermore, the flexibility of processes and regulations within the enterprise necessitates the adaptation of technology to align with these processes (Joseph Jerome et al., 2024; Maroufkhani et al., 2020). At the same time, researchers indicate that the use of and investment in information and communication technologies lead to improved organizational flexibility (Hempell and Zwick, 2005; Bresnahan et al., 2002).

Some studies indicate that top leadership commitment, supply chain strategy, information sharing through IT systems, supplier relationships, flexible staffing, and production or logistics infrastructure are essential determinants influencing supply chain flexibility in production enterprises (Thuy, 2023). It can be inferred that these factors are also critical for enhancing operational flexibility in e-commerce companies. Additionally, research highlights other elements that affect flexibility in the e-commerce sector, including organizational culture, human resource skills, technological capabilities, market dynamics, and the capacity to collaborate with other organizations for resource optimization (Bieńkowska and Sikorski, 2016). Management's willingness to embrace change is regarded as a vital aspect of flexibility. This openness facilitates adaptation to competitive environmental shifts, promotes adaptive decision-making, and enhances responsiveness to uncertainties in innovation, as well as to the evolving needs of the organization and its external environment (Krishna and Jain, 2016; Xhafa et al., 2017).

Similarly, delegating decision-making authority enhances organizational flexibility by reducing information processing demands placed on decision-makers, while fostering teamwork, open communication, and collaborative decision-making. This approach facilitates quicker responses to complex tasks, aligns expertise with decision-making requirements, and improves information processing across hierarchies (Ambec and Poitevin, 2016; Carrasco et al., 2016; Parakhina et al., 2019). Another critical factor influencing organizational flexibility mentioned in the literature is employees' openness to change. It enhances adaptability and responsiveness to evolving circumstances within the organization and supports the implementation of flexible HR systems to manage role-related stress (Augustsson et al., 2017; Jaiswal, 2018).

Other studies indicate that an innovative organizational culture and effective leadership are essential for fostering flexibility, particularly in SMEs (Anning-Dorson, 2021; Phillips and Wright, 2009). Moreover, the literature suggests that the availability of financial, storage, and transportation resources – considered knowledge-based resources – directly influences operational flexibility. Organizations with strong resource capabilities are better positioned to adapt to changes and meet customer demands (Lemańska-Majdzik and Okręglicka, 2024).

The literature analysis indicates that:

Supposition 2: The flexibility of the e-commerce company is influenced by several factors, the most significant among them are management capabilities, resource capabilities, and software capabilities.

3. Research output

3.1. Research methods and sample

The research method employed quantitative studies utilizing internet surveys, which were developed in consultation with e-commerce managers. The online electronic questionnaire included questions pertaining to company characteristics, self-assessment of flexibility, and factors influencing the flexibility of e-commerce enterprises. These questions were measured using a five-point Likert scale. This research approach is well-suited for entrepreneurs and top managers operating in the online environment. The survey was conducted among owners and top managers of Polish enterprises in e-commerce industry, with a total of 213 respondents representing various types of enterprises and trades. Owners and managers were invited to participate in the survey through multiple channels, including emails, internet communicators, social media, phone calls, and face-to-face meetings. The active involvement of researchers in several national e-commerce events was particularly beneficial. However, the distribution of the questionnaire relied on a domestic e-commerce database aimed at securing a balanced and sufficient sample of e-commerce businesses. The characteristics of the research sample are presented in Table 1 below. The collected data were analysed using statistical methods, including structural equation modelling (Bagozzi and Yi, 2012).

There is a great diversity visible among the surveyed enterprises. They differ in the number of employees and turnover, geographic coverage, experience, and the number of completed orders in the month. Such differentiation is characteristic of the e-commerce sector and indicates the representativeness of the surveyed sample.

3.2. Flexibility in the investigated sample

The survey asked respondents to self-assess their company's levels of various types of flexibility, categorized into four areas as described in the literature review (2.1). Some of these areas pertain to offer management, while others focus on logistics and human relations; the last one addresses software. Table 2 presents the levels of various types of flexibility in the surveyed enterprises.

The respondents rated assortment flexibility – the ability of the company to easily change the range of goods offered by the e-store – the highest, followed closely by price and margin flexibility, which refers to the ability to rapidly adjust the prices and margins of the products

Table 1
Characteristics of the research sample.

	Number of respondents	Percentage (%)
Size of company		
Micro-enterprises	175	82.2
Small enterprises	29	13.6
Medium-sized enterprises	4	1.9
Large enterprises	5	2.4
Geographic coverage		
Poland	133	62.0
UE	44	20.7
Europe	8	3.8
World	28	13.1
Average monthly number of completed orders		
Under 300	119	55.9
From 300 to 1000	42	19.7
From 1001 to 10000	37	17.4
From 10001 to 50000	10	4.7
More than 5000	3	1.4

Table 2
Level of companies' flexibility.

Types of flexibility	Level of flexibility					
	1 – No flexibility	2 – Low flexibility	3 – Medium flexibility	4 – High flexibility	5 – Very high flexibility	Average rating
Shaping the offer						
Flexibility of the assortment (*Flex_a*)	1	15	36	80	81	**4.06**
Flexibility in selecting suppliers (*Flex_su*)	13	34	62	59	45	**3.42**
Flexibility of prices and margins (*Flex_p*)	2	15	34	73	89	**4.09**
Logistics						
Flexibility of storage space (*Flex_st*)	26	49	55	50	33	**3.07**
Flexibility of delivery services (*Flex_d*)	7	32	51	69	54	**3.62**
Flexibility of the product packaging process (*Flex_pp*)	10	35	55	76	37	**3.44**
Human resources						
Flexibility of the number of employees (*Flex_ne*)	41	68	49	37	18	**2.64**
Flexibility of employees' working time (*Flex_ew*)	31	21	46	59	56	**3.41**
Flexibility of employment conditions (*Flex_ec*)	32	41	61	41	38	**3.05**
Software						
Flexibility of e-store functionality (*Flex_sf*)	6	31	78	59	39	**3.44**
Flexibility of technology (*Flex_t*)	21	86	55	36	15	**2.71**

offered. Approximately 76% of respondents indicated that they experienced "high flexibility" or "very high" flexibility. The lowest ratings were given to technological flexibility – the ability of the enterprise to quickly and easily change its technological solutions (software) – and employee flexibility, which pertains to the ability of the company to adjust the number of employees swiftly. About 50% of the respondents reported experiencing "no flexibility" or "low flexibility" for technological flexibility are likely associated with the high migration costs involved in transitioning to different software solutions. Additionally, the low scores for employee flexibility may be attributed to the fact that over 82% of the surveyed sample consists of micro-enterprises, which typically employ only a few employees. Consequently, there is limited capacity to reduce the workforce (as seen in one-person enterprises) and hiring an additional employee, which results in a significant percentage increase in labour costs.

In another part of the study, respondents assessed the importance of different factors affecting their enterprise's flexibility level. The size of the impact was measured on a five-point Likert scale, where 1 meant irrelevant, 2 meant minor impact, 3 meant moderate impact, 4 meant significant impact, and 5 meant huge impact. The selected factors and their ratings along with the average rating are presented in Table 3. The selection criterion was the appearance in the literature (2.2) and an average rating of at least 3.4, meaning having greater than average influence.

The research has shown that for most of the tested factors, their impact on flexibility is noticeable. Apart from a few factors, the average rating significantly exceeds the value of 3, which means medium impact. Among the surveyed factors, the standout is management's openness to introducing changes, which received an average rating above 4 (4.13), meaning significant impact.

3.3. Structural equation modelling

Structural equation modelling (SEM) is a widely practiced statistical technique that simultaneously models and estimates complex interactions among numerous dependent and independent variables (Hair et al., 2021). SEM differs from other modelling approaches as they test the direct and indirect effects on pre-assumed causal relationships (Fan et al., 2016). It considers directly observed and latent variables.

Before establishing the SEM model, a critical issue concerning the investigated sample was identified – customer perception of an online store. Unlike the other variables discussed earlier, these variables were not obtained through surveys of managers but from online platforms dedicated to rating e-commerce providers based on customer feedback. The data collected by independent online services is considered a reliable approximation of actual customers' perception of the service levels they have experienced. For this study, the two rating websites most frequently used by customers in the country, Ceneo.pl and Opineo.pl, were exploited. Both platforms gathered numerous opinions along with scaled ratings for each company within the investigated sample. While both websites are similar, Ceneo.pl primarily focuses on price comparison, whereas Opineo.pl emphasizes customer reviews; both have the functionality of online store evaluation. According to the literature, customer ratings and online reviews significantly influence purchasing decisions (Kim et al., 2024). Additionally, the findings by Rosillo-Díaz et al. (2024) suggest that the number of reviews, the credibility of the

Table 3
The impact of selected factors on companies' flexibility.

Factors	Size of influence					
	1 – Irrelevant	2 – Minor impact	3 – Moderate impact	4 – Significant impact	5 – Huge impact	Average rating
Management factors						
Management's openness to introducing changes (*Cap_m*)	8	8	26	77	94	**4.13**
Organizational culture (Cap_oc)	16	22	48	66	61	**3.63**
Decision-making delegation *Cap_dd*)	17	29	55	66	46	**3.45**
Resource factors						
Employees' openness to change (*Cap_e*)	27	10	32	72	72	**3.71**
Financial resource availability (*Cap_f*)	8	15	41	79	70	**3.88**
Software factors						
Availability of functionalities (*Cap_af*)	9	9	49	82	64	**3.86**
Software type (*Cap_t*)	11	19	56	75	52	**3.65**
Software delivery model (*Cap_dm*)	24	24	35	69	61	**3.56**

source, and the usefulness of the reviews are aligned with perceived product quality and purchase intention.

These two website rating services, which are the primary evaluators in the studied area, are treated regarded as of customer perception. The data from the e-commerce review platforms associated with each investigated shop was collected and appropriately recalibrated to create a five-point scale. The percentage of positive reviews for each was considered, and the following intervals were assigned point values: 0–50% of positive opinions was converted to the lowest position on the scale, receiving a score of 1, which indicates a very low perception of service quality; a range of 50–70% was assigned a score of 2; 70–85% received a score of 3; 85–95% was designated a score of 4; and finally, 95% or more of positive opinions was assigned the highest score of 5. The number of positive and negative opinions reviews for e-store, along with percentage of customer opinions, has been demonstrated an effective approximation of perception approximation the comparative study of online bookstores (Urban and Buraczyńska, 2017). The data conversion conducted process resulted in a standardized format, enabling the inclusion of data in SEM. The two variables of customer perception are *Cust_1* and *Cust_2*.

One of the primary motivations behind this study is to elucidate the concept of e-commerce flexibility, components and conditions. Specifically, it aims to determine whether the flexibility of an online store (variables prefixed with *Flex_)* genuinely contributes to enhancing the customer perception of the store's services (latent variable *Cust*). Additionally, the study seeks to identify which company factors (variables prefixed with *Cap_*) facilitate the development of such flexibility in customer perception. The foundational equation is presented below.

*Cap_* → *Flex_* → *Cust*

Based on this general concept, several attempts have been made to develop a satisfactory SEM that meets the requirements for statistical credibility. Ultimately, an SEM with adequate statistical parameters was established. The SEM graph, along with relevant statistics, is presented in Fig. 1 below. However, before that, Tables 4–6 present the statistical data related to the model. The calculations were performed using the IBM SPSS statistical software package, specifically the Amos tool.

As presented in Table 4, the minimum credible requirements are met. The chi-square statistic, which tests the model's fit to the data, is relatively high but still acceptable. A p-value that is very close to zero indicates the statistical significance of the model. Table 5 includes the goodness-of-fit measures for the structural model. The chi-square statistic (CMIN), when comparing the default model with the independent model, supports the acceptance of the proposed SEM. The p-value reflects a very good level of significance.

The regression weights presented in Table 6 illustrate the strength and direction of the influence between variables, which are essential for demonstrating model validity. The estimated regression coefficients indicate a positive direction of influence for each pair. Although the coefficient values are relatively low, the p-values are acceptable across all pairs. The statistics included in Tables 4–6 support the acceptance of this structural model as credible for further detailed analysis. Despite exhibiting moderate or even low levels of statistical relationship strength, the model may provide valuable insights into flexibility issues within the e-commerce organization.

The path analysis diagram in Fig. 1 illustrates a structural model of the relationships derived from survey data combined with data from e-commerce review platforms. One of the variables is latent (*Cust*), while the remaining variables are observable. The numerical values on the arrows in Fig. 1 represent the path coefficients, which indicate the strength of the influence of one variable on another. Meanwhile, the values displayed over the shapes represent the determination coefficients, indicating the percentage of variance explained by the influencing variables.

3.4. Discussion

Among the various types of e-commerce flexibility examined in this study, only two emerged as significant factors influencing customer perception (Cust). The first is assortment flexibility (*Flex_a*), and the second is price and margin flexibility (*Flex_p*). According to the developed structural model, both factors exhibit a comparable strength of influence, with values of .32 and .35, respectively. However, other studies suggest that customer perception is also affected by additional factors, such as reviews from published influencers, which were not included in this research (Suri et al., 2023).

Statistical calculation has demonstrated that only the flexibility of offerings plays a significant role in the SEM equation. The ability to adapt and shape offerings can have a more direct and noticeable impact on customers' perception. This is primarily because customers make purchases to fulfil specific needs for products with particular characteristics. Factors such as product availability, features, and terms of sale significantly influence customer interest and their purchasing decisions from selected online stores. It is important to note that companies that exhibit flexibility in their offerings can respond more swiftly to market changes and better address evolving consumer needs and preferences. Consequently, these companies enhance customers' perception of the service level they receive. The suggestion that flexibility in offerings improves customer satisfaction and affects their product considerations in e-commerce has been discussed in the literature. Research has confirmed that perceived assortment and customization significantly influence customers' perception and decision-making processes (Godek and Eveland, 2018).

At the same time, there is a lack of flexibility in the areas of logistics, human resources, and software within the SEM diagram. Meanwhile, studies confirm the relationship among components of logistics flexibility and its impact on customer satisfaction, indicating that flexible logistics competencies, as well as company capabilities, are crucial for generating customer satisfaction (Maldonado-Guzmán et al., 2017; Shah and Sharma, 2014). Moreover, organizational flexibility in human

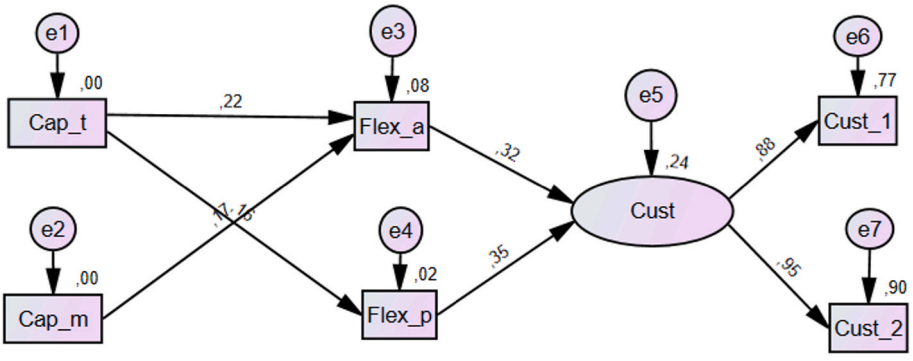


Fig. 1. The SEM paths.

Table 4
Key model's results.

Minimum was achieved
Chi-square = 71.315
Degrees of freedom = 9
Probability level = .000

Table 5
Model fit summary.

Model	NPAR	CMIN	DF	P	CMIN/DF
Default model	18	71.315	9	.000	7.924
Saturated model	27	0	0		
Independence model	6	162.964	21	.000	7.76

Table 6
Regression weights.

			Estimate	S.E.	C.R.	P
Flex_a	<—	Cap_t	.188	.055	3.396	***
Flex_p	<—	Cap_t	.139	.060	2.327	.020
Flex_a	<—	Cap_m	.150	.060	2.502	.012
Cust	<—	Flex_a	.163	.058	2.839	.005
Cust	<—	Flex_p	.169	.055	3.073	.002
Cus_1	<—	Cust	1.000			
Cus_2	<—	Cust	1.000			

resources enhances employee commitment, which, in turn, positively influences customer perception and satisfaction by improving service delivery and responsiveness to consumer needs (Pascua and Japos, 2023; Stanimirov and Zhechev, 2013; Tomar and Dhiman, 2012). Additionally, research shows that flexibility software enhances customer satisfaction by improving service accessibility, transaction speed, navigation comfort, and ease of use. This, in turn, fosters a positive perception of the telecommunication services provided and positively influences customer satisfaction in e-commerce experiences (Mahmoudi et al., 2018; Purwanto and Kuswandi, 2017). Although the impact of other forms of flexibility on customer opinions was not observed, they can affect other aspects of the company's operations. Therefore, while their direct impact on customer perception may not be evident, they can be significant for operational efficiency or cost management, which were not included in the modelling.

Furthermore, it has been demonstrated that assortment flexibility (*Flex_a*) and price and margin flexibility (*Flex_p*) significantly influence customer perception of the store's services (*Cust*). There is a slight difference in the values of the indicators associated with the observable variables *Cust_1* and *Cust_2*. This discrepancy is linked to distinct groups of customers who evaluate their purchases on Ceneo.pl and Opineo.pl. The *Cust_1* variable originates from the Ceneo.pl service, which is utilized to search for the most affordable product offers. Ratings for the online store on this platform are provided by individuals for whom price is often the primary criterion for selecting a shopping place. On Ceneo.pl, all opinions are verified, meaning they pertain to confirmed purchases. In contrast, on Opineo.pl, users seek information regarding the quality of services or products offered. Consequently, ratings are frequently submitted by customers who prioritize quality over price when choosing where to shop. The slightly higher strength of influence indicator for the *Cust_2* variable compared to Cust_1 may be attributed to the fact that Opineo.pl users tend to be more discerning regarding flexibility, and the platform is predominantly populated by stores that offer products more easily tailored to customer expectations. It is likely that Opineo.pl users, as consumers, place a higher value on product quality than on price.

The structural equation demonstrated that management's openness to introducing changes (Cap_m) significantly influences assortment flexibility (Flex_a). This finding aligns with earlier studies that identified management openness as a key factor in determining flexibility (Krishna and Jain, 2016; Xhafa et al., 2017). Furthermore, SEM revealed the impact of software type (*Cap_t*) on both assortment flexibility (*Flex_a*) and price and margin flexibility (*Flex_p*). Previous research has also suggested a relationship between software type and flexibility (Bresnahan et al., 2002; Hempell and Zwick, 2005; Joseph Jerome et al., 2024; Maroufkhani et al., 2020). Both relationships confirm the existence of an influence relationship between these factors and the level of flexibility, as suggested in the literature (Jin and Oriaku, 2013). However, model proven by this study clarifies the types of flexibility within e-commerce enterprises, which are significantly influenced by managerial attitudes and the type of software technology employed.

The model developed here places e-commerce flexibility at its core; it points out a keystone for enhancing customers' perception of online services. The model should be discussed in the context of recent changes within the industry, along with the latest scientific reports. One of the most recent and groundbreaking technologies is artificial intelligence (AI). According to Bilal et al. (2024), AI is revolutionizing consumer–provider interactions by changing the nature of online purchases. Their study finds that AI positively affects consumer experience and consumer engagement on companies' social media. At the same time, the authors highlight the significance of offer flexibility and the pivotal role of technology in enhancing the perceived quality of customer service. These findings make it highly advisable to include AI-based systems in the technological component (*Cap_t*) of the model. In e-commerce, AI-based systems might be crucial for the dynamic modification of offers and pricing to more effectively cater to the changing needs of consumers, thereby enhancing their shopping experiences(Molinillo et al., 2022). Consequently, organizational flexibility that facilitates rapid adjustment of offerings, such as assortment flexibility (*Flex_a*) and price and margin flexibility (*Flex_p*), might be noticeably enhanced by AI-based systems.

Technologies seem to play an increasingly important role in the entire retail and distribution sector, including e-commerce. Zhang (2025) investigated inter-organizational information and communication technology in enhancing e-commerce logistics performance. According to her study, organizational flexibility was found to strengthen the impact of information and communication technology on performance metrics. The study concludes that the necessity of ICT adoption should be strategically integrated with flexible organizational practices. The study supports the model developed here. Companies have to struggle to implement ICT systems promptly, keeping in mind the strategic company's development through greater inter- and intra-organizational flexibility. However, as this model suggests, the flexibility of assortment (Flex_a), along with the price and margin flexibility (Flex_p), is of primary importance.

This study takes a thorough e-commerce business organization perspective. It is interesting that another study investigating the other side, that is, regular online grocery shoppers, brings very similar observations. According to the study by Eriksson and Stenius (2023), the most important factors for improving service quality are (1) a well-fitting, broad, and specialized assortment, (2) delivery time and flexibility, and (3) more pricing options for delivery. The authors investigated many more factors, but these appeared as the most important for frequent online shoppers. This investigation (Eriksson and Stenius, 2023) gives more strength to the model developed here. E-commerce companies should strive more to achieve high flexibility in their operations, aiming for faster and better assortment fit, and

dynamically adjusted prices.

4. Conclusions

The model demonstrates that the flexibility of e-commerce enterprises enhances customer perception. Specifically, increasing an e-commerce company's flexibility in terms of product assortment, pricing and profit margins positively influences customers' perception of its services. This indicates that Supposition 1 has been validated. Additionally, the hypothesis that the flexibility level of an e-commerce company is influenced by various factors (Supposition 2) has also been positively confirmed. The most significant factors affecting assortment flexibility and pricing and margin flexibility are management's willingness to embrace change (management factor) and the type of software utilized (software factor). In conclusion, it can be stated that the key to customer satisfaction lies in the management team's openness to change and the implementation of software that aligns with customer expectations.

4.1. Theoretical and practical implications

The research made a significant contribution to the literature by clarifying how the type of flexibility exhibited by e-commerce companies affects customers' perception of the service they experience. The study identified two types of flexibility (*Flex_*) with a statistically significant impact on service perception variables (*Cust_*). Furthermore, it identified factors that influence the level of flexibility associated with shaping the offerings. The study demonstrated statistically significant relationships between various factors (*Cap_*) and types of flexibility (*Flex_*). This research underscores the importance of assessing and developing enterprise flexibility as a key factor in aiming at superior customer perception. It provides a nuanced perspective on flexibility as a multidimensional concept, adaptable to specific customer priorities such as product availability and pricing competitiveness.

Besides, the validity of simultaneously using data obtained from respondents' ratings and data collected from independent websites in research has been confirmed. This methodological contribution underscores the reliability of combining direct customer feedback with independent platform evaluations for triangulating insights into enterprise performance. Furthermore, the validity of utilizing indicators from external evaluations of online stores, specifically from the Cenoe.pl and Opineo.pl websites, to assess the performance of the enterprise – similar to the customer feedback metrics (CFMs) discussed in the literature (Agag et al., 2023) – has been demonstrated. This highlights the usefulness of actual e-commerce performance indicators in theoretical exploration.

The practical implications of this research for e-commerce involve implementing strategies that enhance assortment flexibility as well as price and margin flexibility. Both types of flexibility can be more effectively developed when a company utilizes software that facilitates the management of a diverse range of products and margins, while also supporting promotions, discounts, and rebates for various customer groups. This underscores the importance of selecting the right software for an e-commerce business – specifically, software that enables rapid price adjustments based on various factors such as purchase volume, customer loyalty, inventory levels, product expiration dates, and the outcomes of negotiations with customers. Additionally, the software may offer functionalities that allow for the ordering of currently unavailable products or for peer-to-peer offerings, which tailor product features to meet customer expectations in a "made to measure" formula.

The flexibility of the company's product assortment can be enhanced by the management's willingness to collaborate with multiple suppliers and to replace them as needed. Additionally, employing various logistics models – such as warehousing, just-in-time delivery, and dropshipping – will enable the company to offer a diverse range of products while minimizing the risks associated with inventory storage. Both types of flexibility necessitate a thorough understanding of customer needs and expectations, which will inform the company's strategies for increasing adaptability. Furthermore, the company should stay abreast of current software trends, including the integration of AI and machine learning for customer personalization, and test its applications in commerce. It is also essential to consider the security of online services, which should be a top priority. Ensuring data protection and privacy will help build and maintain customer trust, a critical factor in sustaining long-term relationships in the e-commerce sector.

From this study emerges a view that to achieve long-term success in e-commerce, companies must deploy adaptive strategies with high priority on technologies that facilitate swift responsiveness to evolving consumer demands. Organizational flexibility, underpinned by AI and other advanced technological innovations, establishes a robust foundation for enhancing service quality, cultivating customer loyalty along with their increasing engagement. Both suppositions presented in the literature section have been confirmed. However, referring to the second one (Supposition 2), two out of three capabilities appeared to be particularly important: management capabilities and software capabilities.

4.2. Limitations and future research

The authors acknowledge several limitations in their research and the resulting model, primarily related to the assumptions made. As previously mentioned, only a select number of factors influencing flexibility were investigated, and not all elements that shape customer perception were examined. Furthermore, the research sample comprised only selected Polish enterprises, leaving it uncertain whether the same relationships exist for businesses in other countries. Similarly, the ratings of e-commerce shops were provided by Polish customers in the Polish language, and it is unclear whether international customers share the same expectations or would evaluate their experiences similarly (Rui et al., 2025; Zielke and Komor, 2025).

Another limitation of the conducted research is the fact that the assessment of customer perception was based on ratings collected on opinion-forming portals, which are not universally utilized by all online stores. Moreover, not all customers provide evaluations of their purchases on these platforms. Other studies have shown that customers are more likely to express negative than positive opinions about their purchases (Barta et al., 2023; Zhang et al., 2024) or post fake reviews (Salminen et al., 2022).

It is advisable to conduct further research to address some of the limitations described. For instance, expanding the research group to include foreign companies could provide valuable insights. To assess the consumer perception, it is recommended to utilize international opinion-collecting websites or to develop a new research tool for conducting interviews with customers. Additionally, consideration should be given to identifying factors that have a limited dependence on the e-commerce company but can significantly impact its flexibility in shaping its offerings.

CRediT authorship contribution statement

Wieslaw Urban: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Barbara Buraczyńska:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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